

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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Prepared By: 3rd Year IT Student

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, definitive set of technical and functional specifications for **Payal's Bakery Cottage**. This document serves as the primary contract and guide for software developers, testers, project managers, and system architects building and validating the system.

1.2 Scope

The software system is a production-grade e-commerce and bakery management system.

- **In Scope:** Customer registration and JWT authentication, product catalog browsing, interactive custom cake configuration engine, shopping cart, delivery date/slot picker, dynamic delivery fees calculation, automated PDF invoice creation, manual UPI payment proof upload and verification flow, admin kitchen priority queue, admin delivery scheduling, inventory reorder tracking, business hours control, and holiday calendar management.
- **Out of Scope:** Native mobile apps (iOS/Android), multi-vendor marketplace functionality, real-time GPS vehicle tracking, shipping outside the 15 km local radius, and WebGL 3D interactive model customizers.

1.3 Definitions, Acronyms and Abbreviations

- **SRS:** Software Requirements Specification
- **PRD:** Product Requirements Document
- **JWT:** JSON Web Token (used for stateless API authorization)
- **RBAC:** Role-Based Access Control
- **OTP:** One-Time Password
- **UPI:** Unified Payments Interface
- **CoD:** Cash on Delivery
- **DTO:** Data Transfer Object
- **UTR:** Unique Transaction Reference (12-digit UPI Transaction ID)

1.4 References

1. *Payal's Bakery Cottage - System Architecture Blueprint v1.0.0*
2. *Payal's Bakery Cottage - Product Requirements Document (PRD) v1.0.0*
3. *Payal's Bakery Cottage - System Readme & Status Overview*
4. *Payal's Bakery Cottage - Screen Documentation & Navigation Flows*
5. *SRS Requirement Specification Template Standard*

1.5 Document Overview

This SRS follows standard engineering documentation structure: Section 2 presents system overview and user roles; Section 3 encapsulates structured functional requirements using standard itemized templates; Section 4 defines non-functional requirements; Section 5 specifies external interface interfaces; Section 6 details system models and diagrams; Section 7 details data structures; Section 8 establishes business logic constraints; Section 9 establishes project acceptance criteria; and Section 10 supplies appendices.

2. Overall Description

2.1 Product Perspective

Payal's Bakery Cottage utilizes a hybrid architectural style combining a layered approach with feature-based modularization. The frontend single-page application connects to a containerized, stateless Java Spring Boot backend via REST APIs over HTTPS. System persistence is backed by MySQL 8, with Redis serving high-speed data caching and Cloud nary providing cloud media storage.

2.2 Product Functions

- **User Authentication & Profiles:** Account creation with email OTP verification, JWT login, profile dashboard, and address book.
- **Product Catalog:** Paginated browsing, multi-attribute filtering (dietary, price, rating), fuzzy search, dynamic stock indicators, and review moderation.
- **Cake Customization Engine:** Step-by-step flavor, weight, shape, message, and reference photo configurator with real-time dynamic pricing.
- **Checkout & Payments:** Preferred delivery/pickup options picker, occasion selector, distance slab fee computation, 3-minute high-resolution UPI QR code generator, screenshot upload with mandatory 12-digit UTR input, and CoD toggle.
- **Admin Operations Dashboard:** Manual payment screenshot and UTR verifier, order queue, kitchen scheduling priority board, inventory reorder threshold monitoring, business hours editor, and holiday blackout manager.

2.3 User Classes and Characteristics

- **Guest User:** Unauthenticated user who can browse the product catalog, filter items, and search products.
- **Customer:** Authenticated user capable of managing saved addresses, configuring custom cakes, placing orders, uploading payment evidence, tracking delivery status, and downloading PDF invoices.
- **Bakery Owner / Head Baker:** Administrative user overseeing production, kitchen queues, preparation schedules, stock levels, business hours, and holiday calendars.
- **Super Admin:** Administrative power user managing user privileges, approving/rejecting manual payment verifications, configuring delivery fees, toggling system-wide CoD, and reviewing audit logs.

2.4 Operating Environment

- **Frontend Runtime:** Web browsers (Chrome 110+, Firefox 115+, Safari 16+, Edge) via React 19 single-page deployment hosted on Vercel CDN.

- **Backend Engine:** Java 21 JRE, Spring Boot 3 running inside containerized Docker instances hosted on Railway/Render.
- **Database & Cache:** Managed MySQL 8 DB instance and Redis Cache Server.
- **External APIs:** Cloudinary (Media upload), Google Maps API (Autocomplete and distance lookups), SMTP Server (Email notifications).

2.5 Design and Implementation Constraints

- **Geographic Limit:** Strictly restricted to a 15-kilometer operational radius from the central bakery.
- **Single Currency:** Transactions are strictly limited to INR (₹).
- **File Upload Constraints:** Maximum 5MB per upload, locked to .png, .jpg, .jpeg formats, uploaded directly to Cloudinary.
- **Timer Bounds:** UPI QR Code active validity window strictly capped at exactly 3 minutes (180 seconds).

2.6 Assumptions and Dependencies

- **Third-Party Services:** High availability of Cloudinary, Google Maps API, and SMTP email services.
- **Network Infrastructure:** End users possess an active internet connection capable of supporting HTTPS REST exchanges and loading static bundle assets.

3. Functional Requirements

3.1 Account & Profile Management

Field	Details
Requirement ID	FR-001
Requirement Name	Customer Account Registration & OTP Verification
Description	Allows new customers to register an account and verify their email via OTP.

Field	Details
Priority	Critical
Preconditions	User is not logged in and is on the registration screen.
Inputs	Name, Email address, Phone number, Password.
Processing	Validate inputs; generate 6-digit OTP; hash password with BCrypt; send verification email.
Outputs	OTP Modal prompt; user status set to Pending Verification until verified.
Exceptions	Duplicate email address returns HTTP 409 Conflict error.
Acceptance Criteria	User receives OTP email within 30 seconds; correct OTP activates account.
Dependencies	Mail Server / SMTP Integration.

3.2 Cake Customization Engine

Field	Details
Requirement ID	FR-002
Requirement Name	Interactive Cake Configurator
Description	Enables customers to configure dynamic custom cake choices and upload photos.
Priority	Critical
Preconditions	User is viewing a customizable product page.

Field	Details
Inputs	Flavor, Weight (0.5kg-5kg), Shape, Egg/Eggless toggle, Custom Text, Reference Photos (up to 3 files), Allergen Checkbox.
Processing	Dynamically adjust pricing based on options; validate lead time gate; execute file signature verification and upload reference photos to Cloudinary.
Outputs	Custom cake object added to cart session with image URLs and calculated total price.
Exceptions	Selected delivery date violates product minimum lead time notice (e.g. <24h).
Acceptance Criteria	Dynamic price updates instantly on option toggle; un-ticked allergen box disables add-to-cart button.
Dependencies	Cloudinary API.

3.3 Dynamic UPI QR Checkout & Payment Submission

Field	Details
Requirement ID	FR-003
Requirement Name	UPI Payment Generation & Proof Submission
Description	Generates high-res unblurred QR Code with 3-min timer; collects screenshot and 12-digit UTR.
Priority	Critical
Preconditions	Cart contains items and user is on Checkout Step 2.

Field	Details
Inputs	Payment method choice (UPI), Payment screenshot file, 12-digit mandatory UPI Transaction ID.
Processing	Generate unblurred UPI QR Code; run 3-minute countdown; process screenshot upload; enforce 12-digit numeric validation on UTR.
Outputs	Order created with status Pending Verification; confirmation email sent.
Exceptions	Timer reaches 0:00 before submission; form fields lock requiring QR regeneration.
Acceptance Criteria	Submission without 12-digit UTR is blocked; expired timer disables submit button.
Dependencies	Cloudinary API, SMTP Email Service.

3.4 Administrative Payment Verification

Field	Details
Requirement ID	FR-004
Requirement Name	Manual Admin Payment Verification Queue
Description	Allows Admin to compare uploaded payment proof against bank records to approve/reject orders.
Priority	Critical
Preconditions	User logged in with ADMIN or SUPER_ADMIN role; orders exist in Pending Verification.

Field	Details
Inputs	Admin action (Approve or Reject), optional rejection reason note.
Processing	Verify screenshot and UTR against bank records; update payment status to Approved or Rejected; transition order state.
Outputs	Payment status changed; automated notification email dispatched to customer.
Exceptions	Admin rejects payment; order moves to Cancelled and customer is alerted.
Acceptance Criteria	Approved payments immediately reflect on the Kitchen Queue Board.
Dependencies	Order Management Service, Notification Service.

3.5 Administrative Delivery & Priority Scheduling

Field	Details
Requirement ID	FR-005
Requirement Name	Kitchen Queue Priority & Delivery Scheduling Board
Description	Enables Admin to assign preparation priorities, schedule exact delivery slots, and assign staff.
Priority	High
Preconditions	Order payment is verified or CoD order accepted.
Inputs	Exact delivery time slot, preparation priority order, assigned bakery delivery staff member.

Field	Details
Processing	Persist delivery schedule; update order state to Order Accepted; issue schedule email.
Outputs	Kitchen board displays chronological prep queue; status emails sent to customer.
Exceptions	Selected prep window exceeds maximum 15 cakes capacity cap; alert issued.
Acceptance Criteria	Scheduled delivery time and assigned staff appear in customer order tracking view.
Dependencies	Database Delivery Schedule Entity.

4. Non-Functional Requirements

4.1 Performance

- **Page Load Time:** Home and catalog listing pages must achieve a Largest Contentful Paint (LCP) under 2.0 seconds over standard mobile connections.
- **API Latency:** 95% of read operations must resolve in <200ms; write endpoints must resolve in <400ms.
- **Image Optimization:** Dynamic transformation via Cloudinary to deliver auto-compressed WebP format assets.

4.2 Security

- **Authentication & Authorization:** Standard JWT authorization with 15-minute short expiration and 7-day HTTPOnly secure refresh cookies, backed by Role-Based Access Control (RBAC).
- **Data Encryption:** TLS 1.3 transport security enforced; user passwords hashed using BCrypt.
- **Defense Mechanisms:** Parametrized SQL queries via Spring Data JPA to prevent SQL injection; CSP headers for XSS protection; Bucket4j rate limiting set to maximum 100 requests/minute per IP.

4.3 Availability and Reliability

- **System Uptime:** Target 99.9% application uptime.
- **Data Backups:** Automated daily backups configured for the production MySQL database instance.

4.4 Usability and Accessibility

- **WCAG Compliance:** Adherence to WCAG 2.1 Level AA guidelines, including appropriate aria-label attributes and keyboard focus management.
- **Responsive Layout:** Tailwind CSS mobile-first breakpoint design catering to viewports from 320px to 2560px.

4.5 Scalability

- **Stateless Backend:** Spring Boot server instances deployed as stateless Docker containers enabling horizontal auto-scaling.
- **Caching Strategy:** Redis caching applied to high-read catalog items, active categories, and coupons to minimize database query strain.

4.6 Maintainability

- **Code Quality:** Backend code coverage $\geq 80\%$ maintained using JUnit 5 and Mockito.

5. External Interface Requirements

5.1 User Interfaces

The web application provides two distinct user interface streams:

1. **Customer Portal:** Landing page, product catalog with bottom drawer filters (mobile), custom cake configuration interface, multi-step cart & checkout wizard, UPI payment screen with timer, and order status tracker.
2. **Back-Office Admin Panel:** Collapsible sidebar navigation, Recharts analytics summary dashboard, product/category CRUD tables, split-panel payment verification board, kitchen queue, and holiday calendar control panel.

5.2 Hardware Interfaces

- The system requires no specialized client hardware. It runs on standard smartphones, tablets, laptops, and desktop computers equipped with modern web browsers.

5.3 Software Interfaces

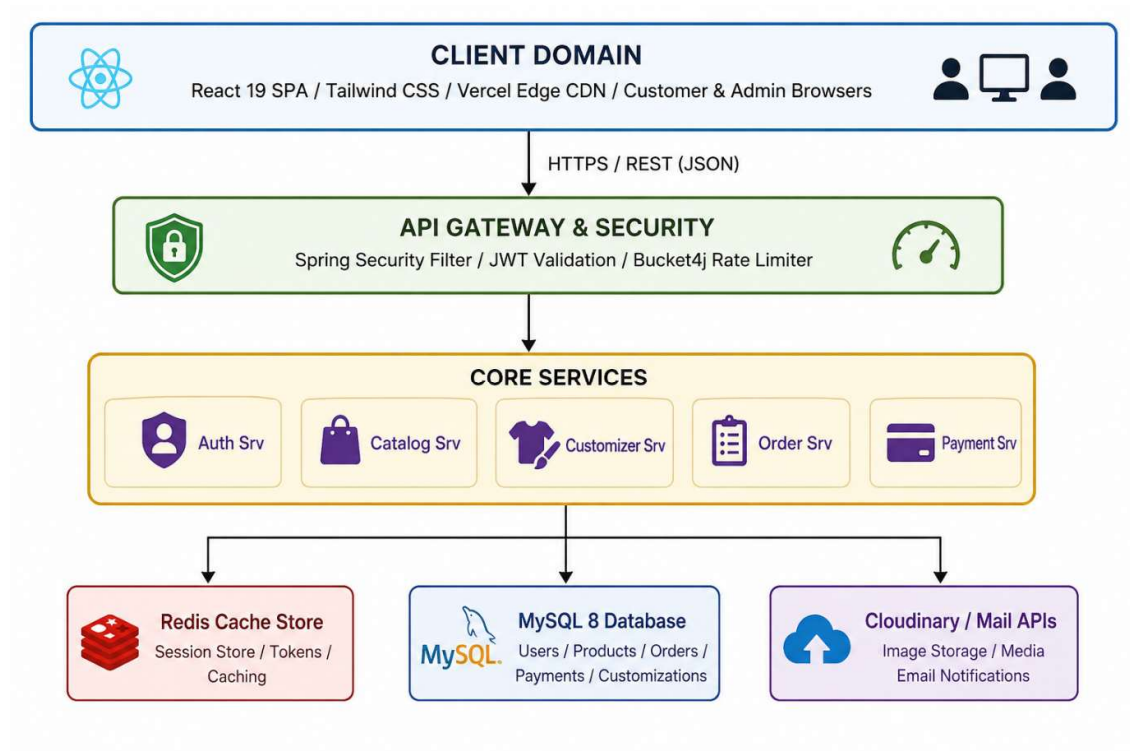
- **Database Engine:** MySQL 8 connected via HikariCP connection pooling.
- **Cache Server:** Redis Cache instance.
- **Cloud Storage API:** Cloudinary SDK for secure signed image uploads.
- **Mapping API:** Google Maps API for address autocomplete and location distance checks.

5.4 Communication Interfaces

- **Transport Protocol:** Standard HTTPS utilizing TLS 1.3 protocol.
- **API Payload:** JSON (JavaScript Object Notation) over HTTP REST endpoints.
- **Notifications:** Asynchronous SMTP protocol for transmitting HTML transaction emails.

6. System Models and Diagrams

6.1 High-Level Architecture Diagram



6.2 Order State Transition Activity Diagram

[*] -> OrderPlaced

OrderPlaced -> PendingPaymentVerification : UPI Selected

OrderPlaced -> PendingAdminAcceptance : CoD Selected & Enabled

PendingPaymentVerification -> PaymentVerified : Admin Approves Payment Proof & UTR

PendingPaymentVerification -> Cancelled : Admin Rejects Payment Proof

PaymentVerified -> PendingAdminAcceptance : Automatic Transition

PendingAdminAcceptance -> OrderAccepted : Admin Accepts & Schedules Delivery Slot

PendingAdminAcceptance -> Cancelled : Admin Rejects Order / Customer Cancels (>24h)

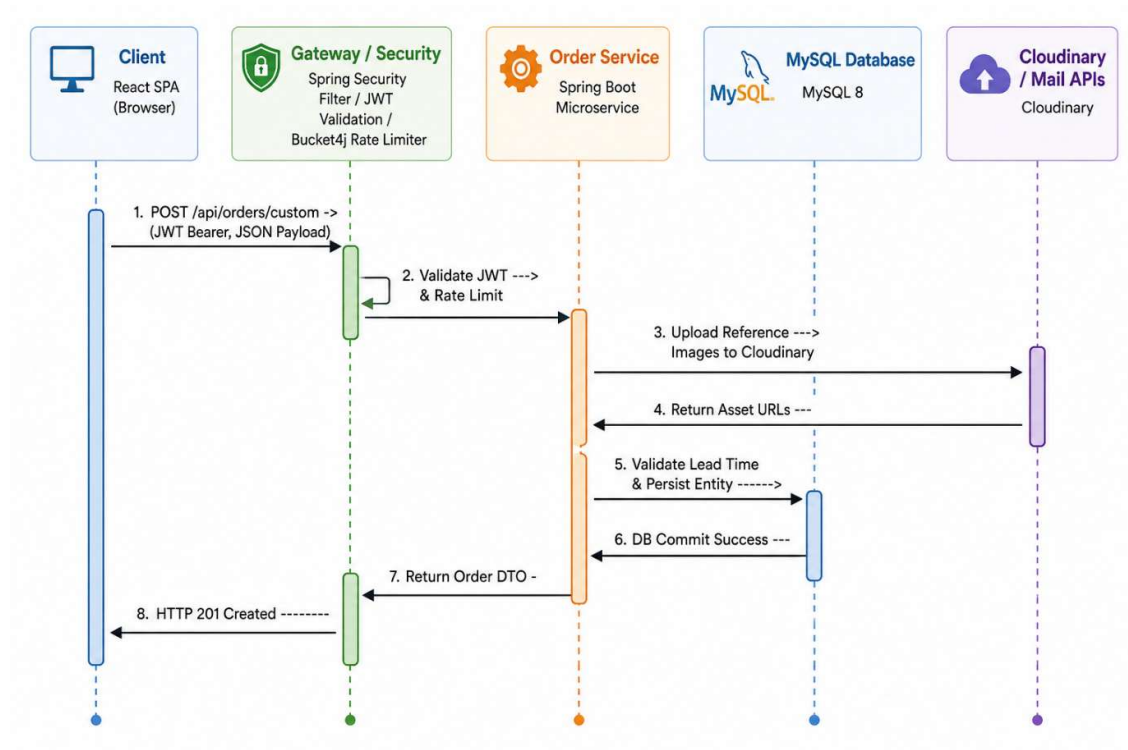
OrderAccepted -> Preparing : Kitchen Starts Baking

Preparing -> OutForDelivery : Admin Dispatches Order

OutForDelivery -> Delivered : Marked Complete by Delivery Staff

Delivered -> [*]

Cancelled -> [*]



7. Data Requirements

7.1 Database Relational Schema & Entities

The application utilizes a production-grade relational database design in MySQL 8 consisting of 18 key entities:

1. **Users:** User ID, Name, Email, Password Hash, Phone, Role (CUSTOMER), Account Verification Status, OTP.
2. **Admins:** Staff ID, Name, Role (BAKERY_OWNER, BAKERY_STAFF, SUPER_ADMIN), Email, Password Hash.
3. **Products:** Product ID, Title, Description, Base Price, Category ID, Availability Flags (AVAILABLE_TODAY, SEASONAL, WEEKEND_SPECIAL, OUT_OF_STOCK), Configurable Minimum Prep Lead Time (12h, 24h, 48h, 72h), Allergen Info, Images.
4. **Categories:** Category ID, Title, Description, Display Priority.
5. **Orders:** Order ID, User ID, Status, Occasion (BIRTHDAY, ANNIVERSARY, WEDDING, etc.), Delivery Option (HOME_DELIVERY, STORE_PICKUP), Preferred Date, Preferred Window, Scheduled Time, Total Amount, Delivery Fee.
6. **Order Items:** Item ID, Order ID, Product ID, Flavor, Weight, Shape, Eggless Toggle, Text Message, Uploaded Image URLs, Subtotal.
7. **Wishlist:** Mapping table associating Users with saved Products.
8. **Cart:** Persistent customer cart items and configuration metadata.
9. **Addresses:** Saved street address lines, Google Maps coordinates, distance from bakery.
10. **Payments:** Payment ID, Order ID, Payment Type (UPI, COD), Amount, Verification Status (PENDING_VERIFICATION, APPROVED, REJECTED), Timestamp.
11. **Payment Proof:** Proof ID, Payment ID, Uploaded Screenshot Cloudinary URL, Mandatory 12-Digit UPI Transaction ID (UTR).
12. **Coupons:** Code, Discount Value, Minimum Order Threshold, Expiry Date, Usage Limits.
13. **Reviews:** Review ID, Product ID, User ID, Rating Stars (1-5), Text Comment, Moderation Status (APPROVED, PENDING).
14. **Inventory:** Ingredient ID, Name, Current Stock Quantity, Reorder Threshold, Expiry Date, Low Stock Warning Flag.
15. **Delivery Schedule:** Schedule ID, Order ID, Assigned Bakery Staff ID, Scheduled Dispatch Time, Actual Delivery Time.

16. **Notifications:** Notification ID, Recipient Email, Event Trigger, Dispatch Status, Timestamp.
17. **Audit Logs:** Log ID, Admin ID, Action Performed, Timestamp, IP Address.
18. **Settings:** Working Hours, Global CoD Toggle, Distance Fee Slab Config, Holiday Date Blocks.

8. Business Rules

1. **Cancellation Constraints:** Custom cake orders cannot be cancelled once accepted by the Admin or when the delivery date is within 24 hours. Standard orders can be cancelled prior to Admin acceptance or if > 24 hours remain before delivery.
2. **Delivery Radius Restriction:** Deliveries are strictly limited to locations within 15 km of the central bakery.
3. **Minimum Preparation Lead Time:** Orders are gated by the product's configured preparation lead time (12, 24, 48, or 72 hours) set by the Admin.
4. **Coupon Rules:** Promotional coupons are single-use, non-stackable, and cannot be combined with loyalty points.
5. **Kitchen Capacity Cap:** Maximum preparation volume is strictly capped at 15 custom cake orders per 2-hour kitchen prep block.
6. **Mandatory Payment Identification:** Payment screenshots submitted without an accompanying 12-digit numeric UPI Transaction ID (UTR) are rejected at the client validation level.
7. **UPI QR Expiry Window:** The checkout QR code remains valid for exactly 3 minutes (180 seconds). Upon timer expiration, screenshot upload and transaction ID inputs are disabled until a new QR Code is generated.
8. **Business Hours Order Queueing:** Orders submitted outside configured business hours (e.g., 9:00 AM – 8:00 PM) are queued for processing on the next working day.
9. **Holiday Date Blocking:** Dates marked as holidays (Festival, Emergency, Maintenance) in the Admin calendar settings are disabled on the user checkout date picker.

9. Acceptance Criteria

Acceptance Criteria 1: Custom Cake Customization Validation

Given a customer is on the custom cake page

When they enter the details:

Field	Value	
Flavor	Premium Red Velvet	
Weight	1.5 kg	
Shape	Heart	
Text on Cake	"Happy Birthday Anna"	
Dietary	Eggless	
Photo	"princess_theme.png" (4MB)	
Delivery	72 hours from current local time	

And they click "Add Custom Cake to Cart"

Then the system updates the dynamic price, uploads the image to Cloudinary, and adds the custom item to the cart session.

Acceptance Criteria 2: UPI QR Code Payment Expiry

Given a customer is on the checkout payment page and selects UPI

When they click "Generate QR Code"

Then a high-resolution unblurred QR Code is displayed with a 3-minute countdown timer.

When the timer reaches 0:00 before proof submission

Then an "Expired" overlay appears, input fields are disabled, and the customer must click "Regenerate QR Code".

When the customer provides a screenshot and a valid 12-digit UTR before expiry

Then the order status transitions to "Pending Verification".

Acceptance Criteria 3: Admin Manual Payment Verification & Delivery Scheduling

Given an Admin is on the Payment Verification dashboard

When they inspect an order with status "Pending Verification", cross-reference the uploaded screenshot and 12-digit UTR, and click "Approve Payment"

Then the payment status updates to "Payment Verified".

When the Admin assigns a preparation slot, schedules the exact delivery time, and assigns a staff member

Then the order status updates to "Order Accepted" and an "Order Accepted" email containing schedule details is sent to the customer.

10. Appendices

10.1 Traceability Matrix

Requirement ID	Description	Source Document Ref	Verification Method
FR-001	Account Registration & Email OTP	PRD §9.1, Screen Docs §2.2-2.3	Integration Test
FR-002	Custom Cake Engine & Photo Upload	PRD §9.3, Screen Docs §2.5	System Test
FR-003	UPI QR Payment & 12-Digit UTR Input	PRD §9.4, Readme §Payment, Screen Docs §2.7	End-to-End Test
FR-004	Admin Manual Payment Verification	PRD §9.5, Readme §Admin, Screen Docs §3.3	User Acceptance Test
FR-005	Kitchen Scheduling & Delivery Priority	PRD §9.5, Readme §Delivery, Screen Docs §3.4	User Acceptance Test

10.2 System Sample Email Notification Workflow

Customer Places Order (UPI Selected)



[Email Dispatched]: "Order Received - Payment Verification Pending"

|



Admin Approves Payment & Schedules Delivery

|



[Email Dispatched]: "Order Accepted & Delivery Scheduled"

|



Kitchen Starts Baking

|



[Email Dispatched]: "Order Preparation Started"

|



Order Dispatched to Delivery Staff

|



[Email Dispatched]: "Out for Delivery"

|



Order Handed to Customer

|



[Email Dispatched]: "Delivered - Thank You for Shopping with Payal's Bakery Cottage!"